About this Book

This book contains selected GATE CE-2 2025 problems and detailed solutions. The problems are presented with the required mathematical derivations, figures, tables, and computational references.

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1. GATE CE-2 2025

A. Formulae

1.1.1 Energy Consumption

- Power-Time-Energy Relation: The total energy consumption E is given by the product of power P, operating time t per day, and the number of days N:

$$E = P \times t \times N \tag{1.1.1}$$

where P is in kilowatts (kW), t is in hours per day, and N is the number of days, yielding energy in kilowatt-hours (kWh).

1.1.2 Alligation Formula

- Mixture Pricing and Proportion: When two varieties of materials or commodities with unit costs C_P and C_Q (where $C_P < CP < C_Q$, with CP being the cost price of the mixture) are mixed in weights W_P and W_Q, the proportion is given by:

$$\frac{W_P}{W_Q} = \frac{C_Q - CP}{CP - C_P} \tag{1.1.2}$$

1.1.3 Heron's Formula and Triangle Area

- Semi-Perimeter: For a triangle with side lengths a, b, and c, the semi-perimeter s is defined as:

$$s = \frac{a + b + c}{2} \tag{1.1.3}$$

- Area via Heron's Formula: The area Δ of the triangle is given by:

$$\Delta = \sqrt{s\,(s-a)\,(s-b)\,(s-c)} \tag{1.1.4}$$

- Area via Base and Altitude: Alternatively, given a base b and corresponding altitude h, the area is:

$$\Delta = \frac{1}{2} \times b \times h \tag{1.1.5}$$

1.1.4 Integration by Parts

- Indefinite Integral Formula: For differentiable functions $u\,(x)$ and $v\,(x)$, the

integration by parts formula is:

$$\int u\,dv = uv - \int v\,du \tag{1.1.6}$$

Expressed explicitly in terms of x:

$$\int u(x)\,v'(x)\;dx = u(x)\,v(x) - \int v(x)\,u'(x)\;dx \tag{1.1.7}$$

1.1.5 Vector Calculus: Curl and Divergence of Velocity Vector

- Two-Dimensional Velocity Field: For a 2D velocity vector $\mathbf{V} = u\hat{i} + v\hat{j}$, where u and v are scalar components along x and y:

  a) z-Component of Curl:

$$(\nabla \times \mathbf{V})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \tag{1.1.8}$$

  b) Divergence of Velocity Vector:

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} \tag{1.1.9}$$

1.1.6 Laplace Transform

- Unilateral Definition: For causal signals starting at $t = 0$:

$$G(s) = \mathcal{L}\{g(t)\} = \int_0^\infty g(t)\,e^{-st}\,dt \tag{1.1.10}$$

- Derivative Properties:

$$y'(t) \overset{\mathcal{L}}{\longleftrightarrow} sY(s) - y(0) \tag{1.1.11}$$

$$y''(t) \overset{\mathcal{L}}{\longleftrightarrow} s^2Y(s) - sy(0) - y'(0) \tag{1.1.12}$$

- Standard Transform Pairs:

$$u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s}, \quad \mathrm{Re}(s) > 0 \tag{1.1.13}$$

$$e^{at}u(t) \overset{\mathcal{L}}{\longleftrightarrow} \frac{1}{s-a}, \quad \mathrm{Re}(s) > a \tag{1.1.14}$$

1.1.7 Higher-Order Derivative Test and Local Extrema

- Critical Points: Points where the first derivative vanishes: $f'(x_0) = 0$.
- Second Derivative Test: Let x_0 be a critical point of $f(x)$:

  a) If $f''(x_0) < 0$, then $f(x)$ has a local maximum at x_0.

b) If $f''(x_0) > 0$, then $f(x)$ has a local minimum at x_0.

c) If $f''(x_0) = 0$, the test is inconclusive, and higher-order derivatives must be evaluated.

1.1.8 Standard Deflection Formulae and Propped Cantilever Analysis

- Deflection due to UDL: The maximum deflection δ_{UDL} at the free end of a cantilever of length L and flexural rigidity EI subjected to a uniformly distributed load of intensity w over the entire span is:

$$\delta_{UDL} = -\frac{wL^4}{8EI} \tag{1.1.15}$$

- Deflection due to End Moment: The maximum deflection δ_M at the free end of a cantilever of length L and flexural rigidity EI subjected to an end moment M is:

$$\delta_M = -\frac{ML^2}{2EI} \tag{1.1.16}$$

- Deflection due to End Load: The deflection δ_R at the free end due to a point load (or redundant reaction R_B) at the tip is:

$$\delta_R = \frac{R_B L^3}{3EI} \tag{1.1.17}$$

- Propped Cantilever Compatibility: For a propped cantilever with a support settlement Δ at the prop (taking upward as positive):

$$\delta_M + \delta_R = -\Delta \implies R_B = \frac{3M}{2L} - \frac{3EI\Delta}{L^3} \tag{1.1.18}$$

1.1.9 Eigenvalues and Eigenvectors of a Matrix

- Characteristic Equation: For a square matrix $\mathbf{A}$, the eigenvalues λ are the roots of the characteristic polynomial:

$$P(\lambda) = \left|\lambda \mathbf{I} - \mathbf{A}\right| = 0 \tag{1.1.19}$$

- Trace and Eigenvalue Sum: The sum of the eigenvalues equals the trace of the matrix:

$$\sum_{i=1}^{n} \lambda_i = \operatorname{tr}(\mathbf{A}) = \sum_{i=1}^{n} a_{ii} \tag{1.1.20}$$

1.1.10 Chemistry and Environmental Engineering Formulae (TOC, TKN, COD)

- Stoichiometric Oxidation Reaction: For an organic compound $C_aH_bO_cN_d$, the complete stoichiometric oxygen demand reaction is:

$$C_aH_bO_cN_d + \left(\frac{4a+b-2c-3d}{4}\right)O_2 \longrightarrow aCO_2 + \frac{b-3d}{2}H_2O + dNH_3 \tag{1.1.21}$$

- Definitions:
  a) **TOC (Total Organic Carbon)**: Measures the total carbon content in the compound ($12a$ per mole).
  b) **TKN (Total Kjeldahl Nitrogen)**: Measures organic and ammonia nitrogen content ($14d$ per mole).
  c) **COD (Chemical Oxygen Demand)**: Measures the oxygen equivalent required for chemical oxidation of the organic matter based on the moles of O_2 consumed ($32 \times$ moles of O_2 per mole of compound).

1.1.11 Hydrostatic Force and Moment about Hinge

- Hydrostatic Force: The total hydrostatic force F on a submerged plane surface of area A with centroidal depth $\bar{y}$ below the free surface is:

$$F = \rho g A \bar{y} \tag{1.1.22}$$

- Moment about Hinge: For a gate or plate pivoted about a hinge, the resultant moment M_P due to linear hydrostatic pressure distribution $p(s) = \rho g \bar{y}(s)$ integrated over the surface length L is:

$$M_P = \int_0^L s\, dF = \rho g b \int_0^L s\left(h - \frac{y(s)}{s}s\right) ds \tag{1.1.23}$$

1.1.12 Highway Engineering Widening Formulae

- Total Extra Widening (W_e): On horizontal curves, extra widening is the sum of mechanical widening (W_m) and psychological widening (W_p):

$$W_e = W_m + W_p \tag{1.1.24}$$

where:

$$W_m = \frac{nl^2}{2R} \quad \text{(Mechanical Widening)} \tag{1.1.25}$$

$$W_p = \frac{V}{9.5\sqrt{R}} \quad \text{(Psychological Widening)} \tag{1.1.26}$$

with n as the number of lanes, l as the wheelbase of the longest vehicle, R as the curve radius, and V as design speed in km/h.

1.1.13 System of Linear Equations and Rouché–Capelli Theorem

- Matrix Form: $\mathbf{Ax} = \mathbf{b}$.
- Solvability via Ranks:
  a) No Solution (Inconsistent): $\text{rank}\,(\mathbf{A}) < \text{rank}\,(\,[\mathbf{A} \mid \mathbf{b}])$
  b) Unique Solution: $\text{rank}\,(\mathbf{A}) = \text{rank}\,(\,[\mathbf{A} \mid \mathbf{b}]) = n$ (number of variables)
  c) Infinitely Many Solutions: $\text{rank}\,(\mathbf{A}) = \text{rank}\,(\,[\mathbf{A} \mid \mathbf{b}]) < n$

1.1.14 Probability and Random Variables

- Mutually Exclusive Events: $\Pr(A \cap B) = 0$ and $\Pr(A \cup B) = \Pr(A) + \Pr(B)$.
- Conditional Probability: $\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)}$ (with $\Pr(B) > 0$). If $B \subset A$, $\Pr(A \mid B) = 1$.
- Mean and Variance of Discrete Random Variables:

$$\mu = E[X] = \sum x_i p_i \tag{1.1.27}$$

$$\sigma^2 = E[X^2] - (E[X])^2 = \sum x_i^2 p_i - \mu^2 \tag{1.1.28}$$

B. Problems

1.2.1 An electricity utility company charges ₹7 per kWh (kilo watt-hour). If a 40-watt desk light is left on for 10 hours each night for 180 days, what would be the cost of energy consumption? If the desk light is on for 2 more hours each night for the 180 days, what would be the percentage-increase in the cost of energy consumption?

(CE2 2025)

a) ₹604.8; 10% b) ₹504; 20% c) ₹604.8; 12% d) ₹720; 15%

Solution: The initial energy consumption is

$$E_1 = P \times t_1 \times N \tag{1.2.1}$$

$$= 0.04\,\text{kW} \times 10\,\text{hours/night} \times 180\,\text{days} \tag{1.2.2}$$

$$= 72\,\text{kWh} \tag{1.2.3}$$

The initial cost at ₹7 per kWh is

$$\text{Initial Cost} = 72\,\text{kWh} \times ₹7/\text{kWh} \tag{1.2.4}$$

$$= ₹504 \tag{1.2.5}$$

With $t_2 = 12$ hours/night, the new energy consumption and cost are

$$E_2 = 0.04\,\text{kW} \times 12\,\text{hours/night} \times 180\,\text{days} \tag{1.2.6}$$

$$= 86.4\,\text{kWh}, \tag{1.2.7}$$

$$\text{New Cost} = 86.4\,\text{kWh} \times ₹7/\text{kWh} \tag{1.2.8}$$

$$= ₹604.8 \tag{1.2.9}$$

Therefore, the percentage increase is

$$\text{Percentage Increase} = \frac{604.8 - 504}{504} \times 100 \tag{1.2.10}$$

$$= 20\%. \tag{1.2.11}$$

Thus, option (B) is correct.

codes/ce2q3/ce2q3.py

1.2.2 A bag contains Violet (V), Yellow (Y), Red (R), and Green (G) balls. On counting them, the following results are obtained:

(i) The sum of Yellow balls and twice the number of Violet balls is 50.

(ii) The sum of Violet and Green balls is 50.

(iii) The sum of Yellow and Red balls is 50.

(iv) The sum of Violet and twice the number of Red balls is 50.

Which one of the following correctly represents the balls in the bag?

(CE2 2025)

a) $V = 10\%$, $R = 20\%$, $Y = 30\%$, $G = 40\%$

b) $V = 40\%$, $R = 10\%$, $Y = 20\%$, $G = 30\%$

c) $V = 30\%$, $R = 40\%$, $Y = 10\%$, $G = 10\%$

d) $V = 20\%$, $R = 30\%$, $Y = 40\%$, $G = 10\%$

Solution: Let the numbers of Violet, Yellow, Red, and Green balls be represented by V, Y, R, G, respectively. The four conditions give

$$\mathbf{A}\mathbf{x} = \mathbf{b}, \tag{1.2.12}$$

where

$$\mathbf{x} = \begin{pmatrix} V \\ Y \\ R \\ G \end{pmatrix}, \qquad \mathbf{A} = \begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{pmatrix}, \qquad \mathbf{b} = \begin{pmatrix} 50 \\ 50 \\ 50 \\ 50 \end{pmatrix}. \tag{1.2.13}$$

The augmented matrix is reduced using elementary row operations. The operations are written directly on the transformation arrows:

$$\begin{pmatrix} 2 & 1 & 0 & 0 & 50 \\ 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 1 & 0 & 50 \\ 1 & 0 & 2 & 0 & 50 \end{pmatrix} \xrightarrow{R_1 \leftrightarrow R_2} \begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 2 & 1 & 0 & 0 & 50 \\ 0 & 1 & 1 & 0 & 50 \\ 1 & 0 & 2 & 0 & 50 \end{pmatrix} \xrightarrow[R_4 \leftarrow R_4 - R_1]{R_2 \leftarrow R_2 - 2R_1} \begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 0 & -2 & -50 \\ 0 & 1 & 1 & 0 & 50 \\ 0 & 0 & 2 & -1 & 0 \end{pmatrix} \tag{1.2.14}$$

Next,

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 0 & -2 & -50 \\ 0 & 1 & 1 & 0 & 50 \\ 0 & 0 & 2 & -1 & 0 \end{pmatrix} \xrightarrow{R_3 \leftarrow R_3 - R_2} \begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 0 & -2 & -50 \\ 0 & 0 & 1 & 2 & 100 \\ 0 & 0 & 2 & -1 & 0 \end{pmatrix} \xrightarrow{R_4 \leftarrow R_4 - 2R_3} \begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 0 & -2 & -50 \\ 0 & 0 & 1 & 2 & 100 \\ 0 & 0 & 0 & -5 & -200 \end{pmatrix} \tag{1.2.15}$$

Finally,

$$\begin{pmatrix} 1 & 0 & 0 & 1 & 50 \\ 0 & 1 & 0 & -2 & -50 \\ 0 & 0 & 1 & 2 & 100 \\ 0 & 0 & 0 & -5 & -200 \end{pmatrix} \xrightarrow[\substack{R_1 \leftarrow R_1 - R_4 \\ R_2 \leftarrow R_2 + 2R_4 \\ R_3 \leftarrow R_3 - 2R_4}]{R_4 \leftarrow -\frac{1}{5} R_4} \begin{pmatrix} 1 & 0 & 0 & 0 & 10 \\ 0 & 1 & 0 & 0 & 30 \\ 0 & 0 & 1 & 0 & 20 \\ 0 & 0 & 0 & 1 & 40 \end{pmatrix}. \tag{1.2.16}$$

Hence,

$$V = 10, \qquad Y = 30, \qquad R = 20, \qquad G = 40. \tag{1.2.17}$$

The total number of balls is

$$T = V + Y + R + G = 100, \tag{1.2.18}$$

$$V = 10\%, \qquad R = 20\%, \qquad Y = 30\%, \qquad G = 40\%. \tag{1.2.19}$$

Hence, option (A) correctly represents the balls in the bag.

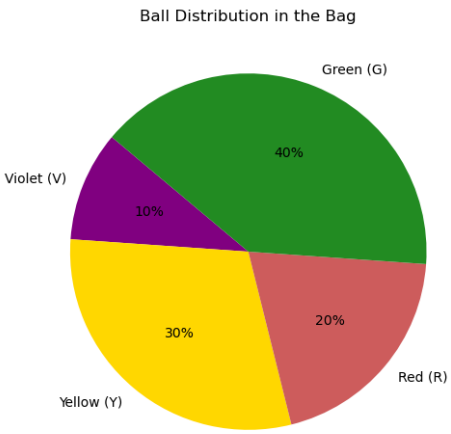


Fig. 1. Pie chart representing ball distribution.

codes/ce2q5/ce2q5.py

1.2.3 Consider a five-digit number PQRST that has distinct digits P, Q, R, S, and T, and satisfies the following conditions: $P < Q$, $S > P > T$, and $R < T$. If integers 1 through 5 are used to construct such a number, the value of P is:

(CE2 2025)

a) 1 b) 2 c) 3 d) 4

Solution: Combining the inequalities gives

$$R < T < P < Q, \qquad T < P < S. \tag{1.2.20}$$

Since $R < T < P$, the digit P must be greater than at least two other distinct digits, namely 1 and 2. Hence $P \neq 1, 2$.

Also, $P < Q$ and $P < S$, so P cannot be the maximum digit 5. Therefore, the only remaining possibility is $P = 3$.

For $P = 3$, the smaller digits $\{1, 2\}$ can be assigned to R and T, while the larger digits $\{4, 5\}$ can be assigned to Q and S. Hence all the inequalities are satisfied.

Thus, option (C) is correct.

codes/ce2q8/ce2q8.c

1.2.4 A business person buys potatoes of two different varieties P and Q, mixes them in a certain ratio and sells them at ₹192 per kg. The cost of variety P is ₹800 for 5 kg. The cost of variety Q is ₹800 for 4 kg. If the person gets 8% profit, what is the $P:Q$ ratio (by weight)?

(CE2 2025)

a) $5:4$ b) $3:4$ c) $3:2$ d) $1:1$

Solution: Let the profit percentage be $p = 0.08$. Let the cost price of the mixture be z. Since $\text{SP} = z(1+p)$,

$$z = \frac{192}{1.08} = \frac{1600}{9}\,₹/\text{kg}, \tag{1.2.21}$$

$$c_P = \frac{800}{5} = 160\,₹/\text{kg}, \qquad c_Q = \frac{800}{4} = 200\,₹/\text{kg}. \tag{1.2.22}$$

Let x kg of variety P and y kg of variety Q be mixed. Equating the total cost to the cost of the mixture gives

$$xc_P + yc_Q = (x+y)\,z, \tag{1.2.23}$$

$$x\,(z - c_P) = y\,(c_Q - z)\,, \tag{1.2.24}$$

$$\frac{x}{y} = \frac{c_Q - z}{z - c_P} \tag{1.2.25}$$

$$= \frac{200 - \frac{1600}{9}}{\frac{1600}{9} - 160} = \frac{5}{4}. \tag{1.2.26}$$

Thus, the required ratio is $5:4$, making option (A) correct.

codes/ce2q9/ce2q9.c

1.2.5 Three villages P, Q, and R are located in such a way that the distance $PQ = 13$ km, $QR = 14$ km, and $RP = 15$ km. A straight road joins Q and R. It is proposed to connect P to this road QR by constructing another road. What is the minimum possible length (in km) of this connecting road?

(CE2 2025)

a) 10.5 b) 11.0 c) 12.0 d) 12.5

Solution: The minimum distance from vertex P to line QR is the altitude h_a. The semi-perimeter of triangle PQR is

$$s = \frac{13 + 14 + 15}{2} = 21. \tag{1.2.27}$$

Using Heron's formula, the area is

$$\Delta = \sqrt{21\,(21-14)\,(21-15)\,(21-13)} \tag{1.2.28}$$

$$= \sqrt{21 \times 7 \times 6 \times 8} \tag{1.2.29}$$

$$= 84\,\text{km}^2. \tag{1.2.30}$$

Taking $QR = 14$ km as the base,

$$h_a = \frac{2\Delta}{14} = \frac{2 \times 84}{14} = 12\,\text{km}. \tag{1.2.31}$$

Thus, option (C) is correct.

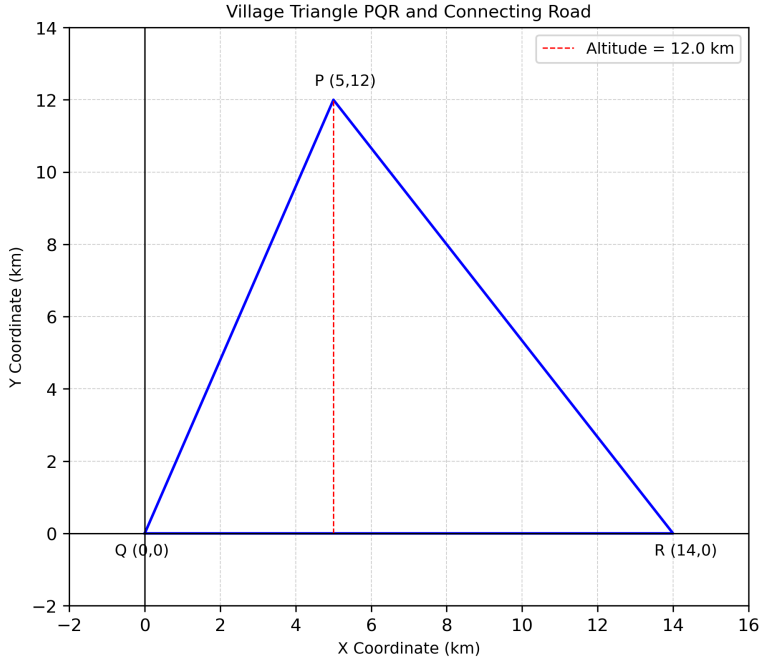


Fig. 2. Triangle PQR geometry and altitude from P to QR.

codes/ce2q10/ce2q10.py

1.2.6 For the matrix $[\mathbf{A}]$ given below, the transpose is:

$$[\mathbf{A}] = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 4 & 5 \\ 4 & 3 & 2 \end{pmatrix} \tag{1.2.32}$$

(CE2 2025)

a) $\begin{pmatrix} 2 & 1 & 4 \\ 3 & 4 & 3 \\ 4 & 5 & 2 \end{pmatrix}$ b) $\begin{pmatrix} 4 & 3 & 2 \\ 5 & 4 & 1 \\ 2 & 3 & 4 \end{pmatrix}$ c) $\begin{pmatrix} 4 & 2 & 3 \\ 5 & 1 & 4 \\ 2 & 4 & 3 \end{pmatrix}$ d) $\begin{pmatrix} 2 & 3 & 4 \\ 1 & 4 & 5 \\ 4 & 3 & 2 \end{pmatrix}$

Solution: The transpose operation interchanges rows and columns. Therefore,

$$[\mathbf{A}^\top]_{ij} = [\mathbf{A}]_{ji}, \tag{1.2.33}$$

$$\mathbf{A}^\top = \begin{pmatrix} 2 & 1 & 4 \\ 3 & 4 & 3 \\ 4 & 5 & 2 \end{pmatrix}. \tag{1.2.34}$$

Thus, option (A) is correct.

codes/ce2q11/ce2q11.py

1.2.7 Integration of $\ln(x)$ with respect to x, i.e., $\int \ln(x)\, dx =$_____

(CE2 2025)

a) $x \ln(x) - x + C$
b) $x - \ln(x) + C$
c) $x \ln(x) + x + C$
d) $\ln(x) - x + C$

Solution: Using integration by parts with $u = \ln(x)$ and $dv = dx$,

$$\int \ln(x)\, dx = x \ln(x) - \int x \cdot \frac{1}{x}\, dx \tag{1.2.35}$$

$$= x \ln(x) - \int 1\, dx \tag{1.2.36}$$

$$= x \ln(x) - x + C. \tag{1.2.37}$$

Thus, option (A) is correct.

codes/ce2q12/ce2q12.py

1.2.8 Consider the pin-jointed truss shown in the figure. Influence line is drawn for the axial force in the member $G - I$, when a unit load travels on the bottom chord of the truss. Identify the CORRECT influence line from the following options.

(CE2 2025)

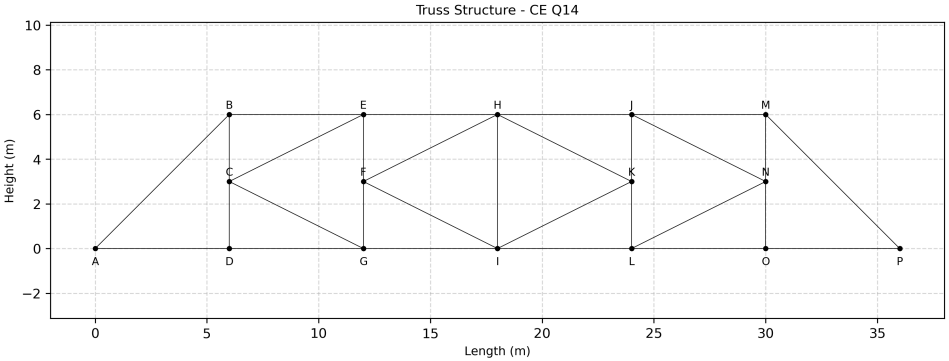


Fig. 3. Pin-jointed truss for Q14.

Solution: The unit load can act at the bottom-chord joints $A, D, G, I, L, O,$ and P. The section-cut analysis gives the following influence ordinates:
Consider a pin-jointed truss of total span L, divided into n equal panels of width d, with a total truss height h. Let x be the variable position of a unit vertical load measured from the left support A.

1. Support Reaction Formulation

Taking moments about the left support A to determine the right support reaction R_P:

$$\sum M_A = 0 \implies R_P \cdot L - 1 \cdot x = 0 \implies R_P = \frac{x}{L}$$

Similarly, the left support reaction R_A is expressed as:

$$R_A = 1 - \frac{x}{L}$$

2. Section-Cut and Moment Equilibrium

Member $G - I$ is a bottom-chord member situated between joint G (at coordinate x_G) and joint I (at coordinate x_I). A vertical section is passed through member $G - I$, and moment equilibrium is considered about the opposing top-chord joint H (located at coordinate x_H).

a) **When the unit load acts to the right of the section ($x \geq x_H$):**
Isolating the portion of the truss to the right of the section and taking moments about joint H:

$$\sum M_H = 0 \implies F_{GI} \cdot h - R_P \cdot (L - x_H) = 0$$

$$\implies F_{GI} = \left(\frac{L - x_H}{h}\right) R_P = \left(\frac{L - x_H}{h \cdot L}\right) x$$

b) **When the unit load acts to the left of the section ($x \leq x_G$):**
Isolating the portion of the truss to the left of the section and taking moments about joint H:

$$\sum M_H = 0 \implies R_A \cdot x_H - 1 \cdot (x_H - x) - F_{GI} \cdot h = 0$$

$$\implies F_{GI} = \frac{R_A \cdot x_H - (x_H - x)}{h}$$

3. General Expression for Joint Positions

For each bottom-chord joint located at position x_i ($i \in \{A, D, G, I, L, O, P\}$), substitute $x = x_i$ into the respective segment's algebraic equation above to obtain the analytical expression for the influence ordinate.

Load at Joints	Tensile value (FGI)
A	0
D	0.33
G	1.33
I	1
L	0.667
O	0.33
P	0

TABLE 1
TENSILE VALUES AT JOINTS

tables/ce2q14/ce2q14.tex

For the member $G-I$, the influence ordinates are positive and vary linearly between consecutive panel points. The maximum ordinate occurs at joint G, while the end-support ordinates are zero.

Therefore, the correct influence line is the positive piecewise-linear line with its peak at G.

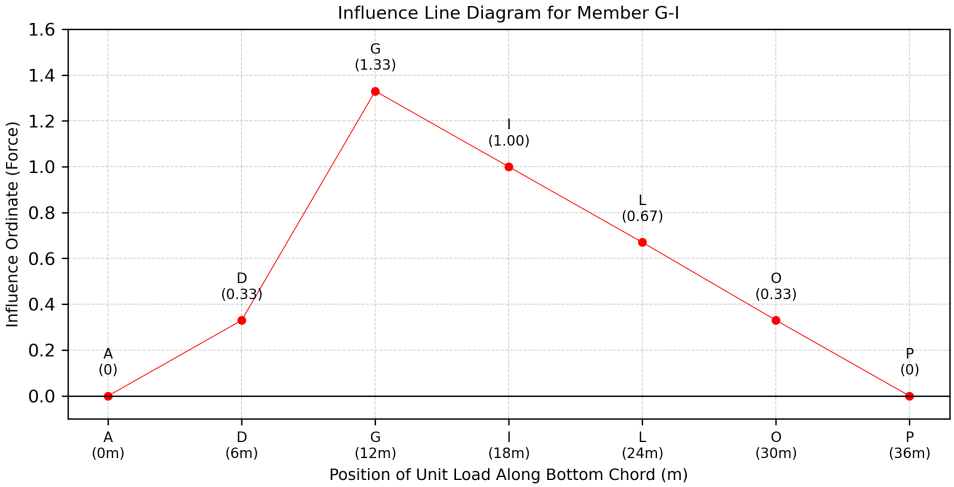


Fig. 4. Influence line graph

Hence, the correct option is (B).

codes/ce2q14/ce2q14.py

1.2.9 Consider a velocity vector $\mathbf{V}$ in (x, y, z) coordinates given by $\mathbf{V} = u\mathbf{x} + v\mathbf{y}$. Pick one or more CORRECT statement(s) from the choices given below.

(CE2 2025)

a) z-component of Curl of velocity: $\nabla \times \mathbf{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}\right)\mathbf{z}$

b) z-component of Curl of velocity: $\nabla \times \mathbf{V} = \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y}\right)\mathbf{z}$

c) Divergence of velocity: $\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$

d) Divergence of velocity: $\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$

Solution: For a two-dimensional velocity field $\mathbf{V} = u\mathbf{x} + v\mathbf{y}$, the z-component of curl is

$$(\nabla \times \mathbf{V})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \tag{1.2.38}$$

Thus, statement (A) is correct.

The divergence is

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}. \tag{1.2.39}$$

Thus, statement (C) is correct.

Therefore, the correct statements are (A) and (C).

codes/ce2q21/ce2q21.py

1.2.10 Given that A and B are not null sets, which of the following statements regarding probability is/are CORRECT?

(CE2 2025)

a) $\Pr(A \cap B) = \Pr(A)\Pr(B)$ if A and B are mutually exclusive.

b) Conditional probability, $\Pr(A \mid B) = 1$ if $B \subset A$.

c) $\Pr(A \cup B) = \Pr(A) + \Pr(B)$, if A and B are mutually exclusive.

d) $\Pr(A \cap B) = 0$, if A and B are independent.

Solution: Since $B \subset A$,

$$A \cap B = B, \tag{1.2.40}$$

$$\Pr(A \mid B) = \frac{\Pr(A \cap B)}{\Pr(B)} = \frac{\Pr(B)}{\Pr(B)} = 1. \tag{1.2.41}$$

Hence, statement (B) is correct.

For mutually exclusive events,

$$\Pr(A \cap B) = 0, \tag{1.2.42}$$

$$\Pr(A \cup B) = \Pr(A) + \Pr(B). \tag{1.2.43}$$

Hence, statement (C) is correct. Statements (A) and (D) are incorrect for non-null sets.

Therefore, the correct statements are (B) and (C).

codes/ce2q22/ce2q22.py

1.2.11 Pick the CORRECT solution for the following differential equation

$$\frac{dy}{dx} = e^{x-y}. \tag{1.2.44}$$

(CE2 2025)

a) $y = \ln\left(e^x + \text{Constant}\right)$

b) $\ln(y) = x + \text{Constant}$

c) $\ln(y) = \ln\left(e^x\right) + \text{Constant}$

d) $y = x + \text{Constant}$

Solution: The nonlinear equation is transformed using $u = e^y$.

$$\text{Given:} \quad \frac{dy}{dx} = e^{x-y} = e^x e^{-y}, \tag{1.2.45}$$

$$\text{Let:} \quad u = e^y, \tag{1.2.46}$$

$$\frac{du}{dx} = e^y \frac{dy}{dx} = e^x, \tag{1.2.47}$$

$$\text{Laplace transform:} \quad \mathcal{L}\left\{\frac{du}{dx}\right\} = sU(s) - u(0), \tag{1.2.48}$$

$$\mathcal{L}\{e^x\} = \frac{1}{s-1}, \tag{1.2.49}$$

$$sU(s) - u(0) = \frac{1}{s-1}, \tag{1.2.50}$$

$$U(s) = \frac{1}{s(s-1)} + \frac{u(0)}{s}, \tag{1.2.51}$$

$$\tag{1.2.52}$$

$$\text{Partial fractions:} \quad \frac{1}{s(s-1)} = -\frac{1}{s} + \frac{1}{s-1}, \tag{1.2.53}$$

$$U(s) = \frac{u(0)-1}{s} + \frac{1}{s-1}, \tag{1.2.54}$$

$$\text{Inverse Laplace:} \quad u(x) = u(0) - 1 + e^x, \tag{1.2.55}$$

$$\text{Since } u = e^y: \quad e^y = e^x + C, \tag{1.2.56}$$

$$\text{Taking Log:} \quad y = \ln\left(e^x + C\right). \tag{1.2.57}$$

Hence, the correct solution is

$$y = \ln\left(e^x + \text{Constant}\right). \tag{1.2.58}$$

Therefore, the correct option is (A).

codes/ce2q36/ce2q36.py

1.2.12 The figure shows a propped cantilever with uniform flexural rigidity EI (in N.m^2) and subjected to a moment M (in N.m). Consider forces and displacements in the upward direction as positive. Find the upward reaction at the propped support B (in N) when this support settles by $(-\Delta)$, given in metres.

(CE2 2025)

a) $\frac{3M}{2L} - \frac{6EI\Delta}{L^3}$ b) $\frac{8M}{3L} - \frac{2EI\Delta}{L^3}$ c) $\frac{3M}{2L} - \frac{3EI\Delta}{L^3}$ d) $\frac{M}{L} - \frac{8EI\Delta}{L^3}$

Solution: Treat the reaction at the prop B as the redundant R_B. The compatibility equations

are

$$\delta_B = -\Delta, \tag{1.2.59}$$

$$\delta_M = -\frac{ML^2}{2EI}, \tag{1.2.60}$$

$$\delta_R = \frac{R_B L^3}{3EI}, \tag{1.2.61}$$

$$\delta_M + \delta_R = -\Delta. \tag{1.2.62}$$

Therefore,

$$-\frac{ML^2}{2EI} + \frac{R_B L^3}{3EI} = -\Delta, \tag{1.2.63}$$

$$-\frac{3M}{2L} + R_B = -\frac{3EI\Delta}{L^3}, \tag{1.2.64}$$

$$R_B = \frac{3M}{2L} - \frac{3EI\Delta}{L^3}. \tag{1.2.65}$$

Thus, the upward reaction at support B is

$$R_B = \frac{3M}{2L} - \frac{3EI\Delta}{L^3}. \tag{1.2.66}$$

Hence, the correct option is (C).

codes/ce2q38/ce2q38.py

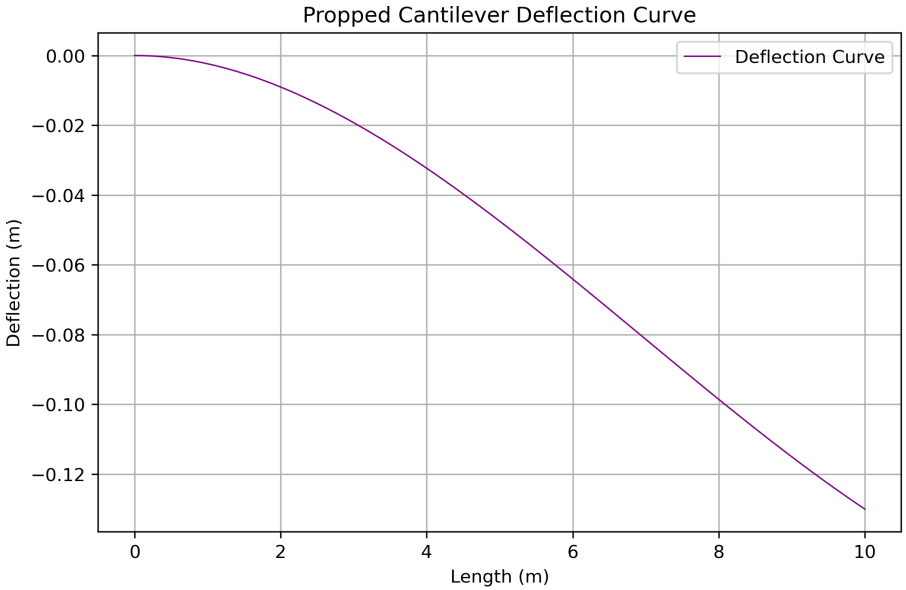


Fig. 5. Propped cantilever and support settlement.

1.2.13 Consider the function given below and pick one or more CORRECT statement(s) from the following choices.

$$f(x) = x^3 - \frac{15}{2}x^2 + 18x + 20. \tag{1.2.67}$$

(CE2 2025)

a) $f(x)$ has a local minimum at $x = 3$.

b) $f(x)$ has a local maximum at $x = 3$.

c) $f(x)$ has a local minimum at $x = 2$.

d) $f(x)$ has a local maximum at $x = 2$.

Solution: The critical points are obtained from $f'(x) = 0$:

$$f'(x) = 3x^2 - 15x + 18 \tag{1.2.68}$$

$$= 3(x^2 - 5x + 6) \tag{1.2.69}$$

$$= 3(x-2)(x-3). \tag{1.2.70}$$

Hence,

$$x = 2, \qquad x = 3. \tag{1.2.71}$$

The second derivative is

$$f''(x) = 6x - 15. \tag{1.2.72}$$

At $x = 2$,

$$f''(2) = 12 - 15 = -3 < 0, \tag{1.2.73}$$

so $x = 2$ is a local maximum.

At $x = 3$,

$$f''(3) = 18 - 15 = 3 > 0, \tag{1.2.74}$$

so $x = 3$ is a local minimum.

Therefore, the correct statements are (A) and (D).

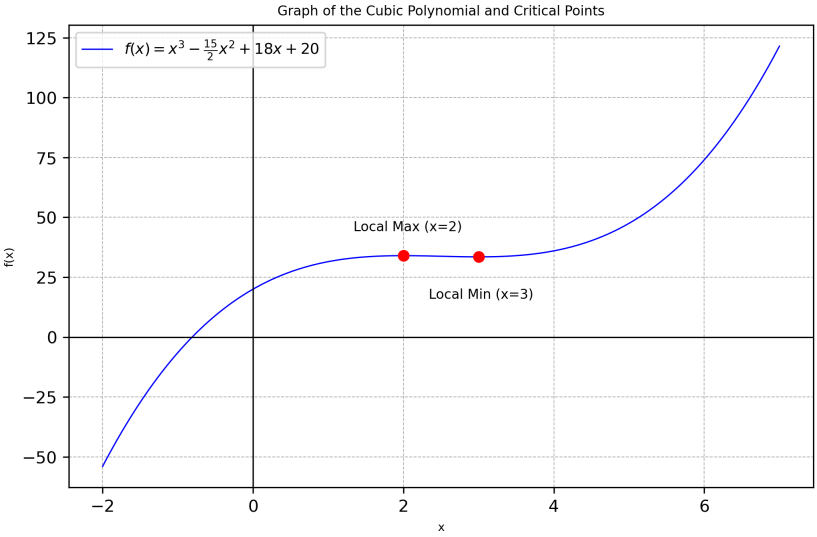


Fig. 6. Graph of the cubic function.

codes/ce2q44/ce2q44.py

1.2.14 Pick the CORRECT eigen value(s) of the matrix $\mathbf{A}$ from the following choices.

$$\mathbf{A} = \begin{pmatrix} 6 & 8 \\ 4 & 2 \end{pmatrix}. \tag{1.2.75}$$

(CE2 2025)

a) 10 b) 4 c) -2 d) -10

Solution: The eigenvalues satisfy the characteristic equation

$$\det(\mathbf{A} - \lambda I) = 0 \tag{1.2.76}$$

$$\det\begin{pmatrix} 6-\lambda & 8 \\ 4 & 2-\lambda \end{pmatrix} = 0 \tag{1.2.77}$$

$$(6-\lambda)(2-\lambda) - 32 = 0 \tag{1.2.78}$$

$$\lambda^2 - 8\lambda - 20 = 0 \tag{1.2.79}$$

$$(\lambda - 10)(\lambda + 2) = 0. \tag{1.2.80}$$

Hence,

$$\lambda = 10, \qquad \lambda = -2. \tag{1.2.81}$$

So, the correct options are (A) and (C).

codes/ce2q45/ce2q45.py

1.2.15 In the pin-jointed truss shown in the figure, the members that carry zero force are identified. Which of the following options is/are zero-force members?

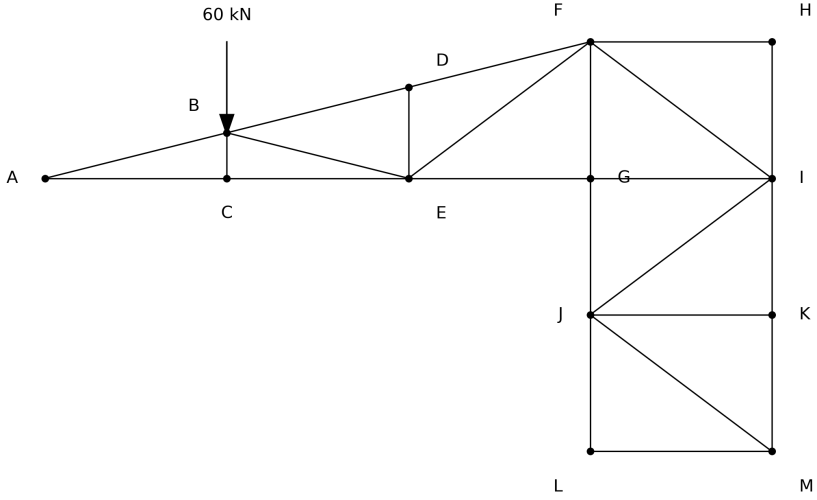


Fig. 7. Truss diagram for identifying zero-force members in Q46.

(CE2 2025)

a) BC b) EG c) FI d) JK

Solution: We apply the standard zero-force member rules for pin-jointed trusses:

a) **At Joint C:** The bottom chord members AC and CE are collinear, and no external load acts at joint C. Therefore, member BC carries zero force:

$$BC = 0 \tag{1.2.82}$$

b) **At Joint J:** The horizontal chord members meeting at joint J are collinear, and no external load or support reaction is applied at joint J. Hence, member JK carries zero force:

$$JK = 0 \tag{1.2.83}$$

Thus, among the choices provided, both BC and JK are zero-force members.
Therefore, the correct options are (A) and (D).

codes/ce2q46/ce2q46.py

1.2.16 A compound has a general formula $C_aH_bO_cN_d$ and molecular weight 187. A 935 mg/l solution of the compound is prepared in distilled deionized water. The Total Organic Carbon (TOC) is measured as 360 mg/l (as C). The Chemical Oxygen Demand (COD) and the Total Kjeldahl Nitrogen (TKN) are determined as 600 mg/l (as O_2) and 140 mg/l (as N), respectively.
As per the reaction

$$C_aH_bO_cN_d + \left(\frac{4a+b-2c-3d}{4}\right) O_2 \rightarrow aCO_2 + \frac{b-3d}{2} H_2O + dNH_3, \tag{1.2.84}$$

which of the following options is/are CORRECT?
Atomic weights: $C = 12$, $H = 1$, $O = 16$, $N = 14$

Quantity	**Value**
Molecular Weight	187
Solution Concentration	935
TOC	360
COD	600
TKN	140

TABLE 2
GIVEN VALUES

tables/ce2q49/ce2q49.tex

(CE2 2025)

a) $a = 6$ b) $b = 7$ c) $c = 5$ d) $d = 3$

Solution: The concentration of the compound is

$$C_{\text{soln}} = 935 \text{ mg/l} = 0.935 \text{ g/l}, \tag{1.2.85}$$

so the molar concentration is

$$C_m = \frac{0.935}{187} = 5 \times 10^{-3} \text{ mol/l}. \tag{1.2.86}$$

From the TOC data,

$$\left(5 \times 10^{-3}\right)(12a) = 360 \times 10^{-3}, \tag{1.2.87}$$

$$0.06a = 0.36, \tag{1.2.88}$$

$$a = 6. \tag{1.2.89}$$

From the TKN data,

$$\left(5 \times 10^{-3}\right)(14d) = 140 \times 10^{-3}, \tag{1.2.90}$$

$$0.07d = 0.14, \tag{1.2.91}$$

$$d = 2. \tag{1.2.92}$$

Hence, option (D) is false.
Using the molecular weight,

$$12a + b + 16c + 14d = 187, \tag{1.2.93}$$

$$72 + b + 16c + 28 = 187, \tag{1.2.94}$$

$$b + 16c = 87. \tag{1.2.95}$$

From the COD data,

$$C_{\text{O}_2} = 600 \text{ mg/l} = 0.6 \text{ g/l}, \tag{1.2.96}$$

and the oxygen consumed per mole of compound is

$$\frac{0.6}{5 \times 10^{-3}} = 120 \text{ g/mol of } O_2, \tag{1.2.97}$$

$$n_{O_2} = \frac{120}{32} = 3.75. \tag{1.2.98}$$

Therefore,

$$\frac{4a+b-2c-3d}{4} = 3.75, \tag{1.2.99}$$

$$18+b-2c = 15, \tag{1.2.100}$$

$$b-2c = -3. \tag{1.2.101}$$

Thus,

$$\begin{pmatrix} 1 & 16 \\ 1 & -2 \end{pmatrix} \begin{pmatrix} b \\ c \end{pmatrix} = \begin{pmatrix} 87 \\ -3 \end{pmatrix}. \tag{1.2.102}$$

Row reduction gives

$$b = 7, \qquad c = 5, \tag{1.2.103}$$

$$a = 6, \qquad d = 2. \tag{1.2.104}$$

Hence,

$$C_aH_bO_cN_d = C_6H_7O_5N_2. \tag{1.2.105}$$

Therefore, the correct options are (A), (B), and (C).

codes/ce2q49/ce2q49.py

1.2.17 Consider a discrete random variable X whose probabilities are given below. The standard deviation of the random variable is _____ (round off to one decimal place).

xi	1	2	4	8
P(X = xi)	0.3	1	0.3	0.3

TABLE 3
PROBABILITY TABLE

tables/ce2q51/ce2q51.tex

(CE2 2025)

Solution: The mean is

$$\mu = E[X] = \sum_i x_i p_i \tag{1.2.106}$$

$$= 1(0.3) + 2(0.1) + 4(0.3) + 8(0.3) = 4.1. \tag{1.2.107}$$

The second moment is

$$E[X^2] = \sum_i x_i^2 p_i \tag{1.2.108}$$

$$= 1^2(0.3) + 2^2(0.1) + 4^2(0.3) + 8^2(0.3) = 24.7. \tag{1.2.109}$$

Hence,

$$\sigma^2 = E[X^2] - \mu^2 = 24.7 - (4.1)^2 = 7.89, \tag{1.2.110}$$

and therefore

$$\sigma = \sqrt{7.89} \approx 2.8. \tag{1.2.111}$$

codes/ce2q51/ce2q51.py

1.2.18 The figure shows a network diagram for a construction project. The activities A, B, C, D, E, and F are represented by arrows and their durations are shown in the figure.
The total float available for the activity E in day(s) is equal to _____ (round off to the nearest integer).

(CE2 2025)

Solution: Let x_i denote the occurrence time of event i and define

$$\mathbf{x} = \begin{pmatrix} x_1 & x_2 & x_3 & x_4 & x_5 \end{pmatrix}^\top. \tag{1.2.112}$$

For an activity from event i to event j with duration d_{ij}, the precedence constraint is

$$x_j - x_i \geq d_{ij}. \tag{1.2.113}$$

For the given network,

$$x_2 - x_1 \geq 3, \qquad x_3 - x_1 \geq 5, \tag{1.2.114}$$

$$x_4 - x_2 \geq 5, \qquad x_4 - x_3 \geq 2, \tag{1.2.115}$$

$$x_5 - x_4 \geq 4, \qquad x_5 - x_3 \geq 8. \tag{1.2.116}$$

These constraints can be written in the matrix form

$$\mathbf{B}\mathbf{x} \geq \mathbf{d}, \tag{1.2.117}$$

where

$$\mathbf{B} = \begin{pmatrix} -1 & 1 & 0 & 0 & 0 \\ -1 & 0 & 1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & -1 & 1 \\ 0 & 0 & -1 & 0 & 1 \end{pmatrix}, \qquad \mathbf{d} = \begin{pmatrix} 3 & 5 & 5 & 2 & 4 & 8 \end{pmatrix}^{\top}. \tag{1.2.118}$$

The project starts at event 1, hence $x_1 = 0$.

The forward-time optimization problem is therefore

$$\begin{aligned} \underset{\mathbf{x}}{\text{minimize}} \quad & x_5 \\ \text{subject to} \quad & \mathbf{B}\mathbf{x} \geq \mathbf{d}, \\ & x_1 = 0. \end{aligned} \tag{1.2.119}$$

The optimal solution is

$$\mathbf{x}^{E} = \begin{pmatrix} 0 & 3 & 5 & 8 & 13 \end{pmatrix}^{\top}, \tag{1.2.120}$$

so that $E_4 = 8$ and $T = E_5 = 13$ days.

For the backward pass, the latest feasible occurrence time of event 4 is

$$x_4^{L} = 9. \tag{1.2.121}$$

For activity $E : 4 \to 5$, with duration $d_E = 4$ days, the latest start time is

$$LST_E = L_5 - d_E = 13 - 4 = 9. \tag{1.2.122}$$

Therefore, the total float is

$$TF_E = LST_E - EST_E = 9 - 8 = 1 \text{ day}. \tag{1.2.123}$$

Hence, the total float available for activity E is

$$\boxed{1}. \tag{1.2.124}$$

codes/ce2q53/ce2q53.py